

A Project Report on
Cognitive Gaming Control Mechanism using EEG

Submitted in partial fulfillment for award of

Bachelor of Technology
Degree
in
Computer Science and Engineering

By

N. Ooha (Y22ACS513)

Sk. Najma (Y22ACS559)

P. Midhun Chakravarthi (Y22ACS517)

Y. Sudha (L22ACS610)



Under the guidance of
Dr. R. Venkata Lakshmi, M.Tech., Ph.D.
Asst. Professor

Department of Computer Science and Engineering

Bapatla Engineering College

(Autonomous)

(Affiliated to Acharya Nagarjuna University)

BAPATLA – 522 102, Andhra Pradesh, INDIA

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**Department of
Computer Science and Engineering**



CERTIFICATE

This is to certify that the project report entitled **Cognitive Gaming Control Mechanism using EEG** that is being submitted by N. Ooha (Y22ACS513), Sk. Najma (Y22ACS559), P. Midhun Chakravarthi (Y22ACS517), Y. Sudha (L22ACS610), in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science & Engineering to the Acharya Nagarjuna University is a record of bonafide work carried out by them under our guidance and supervision.

Date:

Signature of the Guide
Dr. R. Venkata Lakshmi
Assistant Professor

Signature of the HOD
Dr. M. Rajesh Babu
Assoc. Prof. & Head

External Signature

DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

N. Ooha (Y22ACS513)

Sk. Najma (Y22ACS559)

P. Midhun Chakravarthi (Y22ACS517)

Y. Sudha (L22ACS610)

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Students

N. Ooha (Y22ACS513)

Sk. Najma (Y22ACS559)

P. Midhun Chakravarthi (Y22ACS517)

Y. Sudha (L22ACS610)

ABSTRACT

This project presents a cognitive gaming control system based on Brain-Computer Interface (BCI) technology using Electroencephalography (EEG) signals. The main objective is to enable users to control a gaming application using their mental state, reducing the need for traditional input devices such as keyboards or controllers.

The system captures brain signals through electrodes connected to a BIO AMP (EXG pill) and transmits the data to a computer using a microcontroller. The acquired signals are preprocessed using filtering and normalization techniques to remove noise and improve signal quality. Meaningful features are then extracted from the processed data to represent the characteristics of the EEG signals.

A combination of machine learning and deep learning techniques is used to train a model that can classify the user's mental state as attentive or relaxed. The trained model is integrated into a real-time system that continuously processes incoming EEG data and predicts the user's state.

Based on the prediction, control commands are generated to operate a gaming application, enabling hands-free interaction. This system enhances user experience and demonstrates the practical application of EEG-based cognitive control in real-time environments.

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LIST OF ABBREVIATIONS

BCI – Brain Computer Interface
EEG – Electroencephalography
ML – Machine Learning
DL – Deep Learning
ANN – Artificial Neural Network
SRS – Software Requirement Specification
DFD – Data Flow Diagram
UML – Unified Modeling Language
EXG – Electrophysiology Signal
BIO AMP – Biological Amplifier
PSD – Power Spectral Density
CPU – Central Processing Unit
USB – Universal Serial Bus
IDE – Integrated Development Environment
GUI – Graphical User Interface

1. INTRODUCTION

Brain-Computer Interface (BCI) technology is an emerging field that enables direct communication between the human brain and external devices without relying on traditional physical input methods. It works by capturing brain signals, interpreting them, and converting them into commands that can control various systems. This technology creates a new way of interaction between humans and machines, making communication faster and more natural.

One of the most widely used techniques in BCI systems is Electroencephalography (EEG). EEG is a non-invasive method that records the electrical activity of the brain using sensors placed on the scalp. These signals represent different mental states such as attention, relaxation, and cognitive activity. Due to its safety, affordability, and real-time capabilities, EEG has become a popular choice for developing brain-controlled applications.

In recent years, BCI systems have been widely explored in several domains, including healthcare, assistive technologies, education, and entertainment. In the field of gaming, there is a growing interest in developing systems that allow users to interact with games using their mental state rather than physical controllers. This approach not only enhances user engagement but also provides a more immersive and innovative gaming experience.

Cognitive gaming control is one such application where brain signals are used to control game actions. By detecting the user's level of attention or relaxation, the system can respond accordingly, allowing the user to influence the game environment through mental focus. This type of interaction reduces dependency on traditional input devices and opens new possibilities for hands-free control.

Overall, BCI-based systems using EEG signals represent a significant advancement in human-computer interaction. They offer a promising future for developing intelligent, adaptive, and user-friendly systems that can be applied in various real-world scenarios, including gaming, rehabilitation, and assistive technologies.

1.1 MOTIVATION

The main motivation behind this project is to explore a new way of controlling systems without relying on physical input devices. Traditional interaction methods such as keyboards and controllers are widely used, but they may not always provide flexibility or accessibility for all users. This led to the idea of developing a system that operates based on human mental states.

Another important motivation is the increasing demand for interactive and user-centric applications. In gaming, there is a need to move beyond conventional control methods and create experiences that respond directly to the user's level of focus and engagement. This project contributes to that idea by introducing a simple form of cognitive control.

The project is also motivated by the challenges involved in handling real-time data, such as noise removal, feature extraction, and accurate prediction. Addressing these challenges provides valuable understanding in signal processing and machine learning techniques.

Furthermore, the possibility of extending such systems for assistive applications adds significance to this work. A system that can function without physical effort can be useful in situations where traditional interaction methods are limited or not possible.

Overall, the motivation of this project is to develop a system that is not only technically effective but also practical, interactive, and meaningful for real-world applications.

1.2 PROBLEM DEFINITION

The main problem addressed in this project is to develop a system that can accurately identify a user's mental state using EEG signals and use it for real-time control.

EEG signals are highly complex, noisy, and vary from person to person, making them difficult to interpret directly. Extracting meaningful information from these signals and distinguishing between different cognitive states such as attentive and

relaxed is a challenging task. In addition, real-time implementation requires the system to process continuous data quickly and make reliable predictions without delay.

Another challenge is ensuring consistency and accuracy in classification despite variations in signal quality and external disturbances. Without proper handling, these factors can lead to incorrect predictions and unstable system behavior.

Therefore, the problem is to design a robust system that can efficiently process EEG signals, accurately classify the user's mental state, and ensure reliable real-time control despite signal variability and noise.

1.3 OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- a) To develop a system that can detect the user's mental state using EEG signals
- b) To accurately classify cognitive states such as attentive and relaxed
- c) To design a method for processing and analyzing EEG data effectively
- d) To implement a real-time prediction system for continuous monitoring
- e) To control a gaming application based on the user's mental state
- f) To ensure reliable and stable system performance during real-time operation
- g) To explore the practical application of brain-computer interface technology

2. LITERATURE SURVEY

Brain-Computer Interface (BCI) systems have gained significant attention due to their ability to enable direct communication between the human brain and external devices. Among various techniques, Electroencephalography (EEG) is widely used for capturing brain signals because it is non-invasive, cost-effective, and suitable for real-time applications.

Researchers have explored EEG signals to identify different cognitive states such as attention, relaxation, and mental workload. Since EEG signals are highly complex and affected by noise, various preprocessing techniques like filtering and normalization are applied to improve signal quality.

For classification of EEG signals, several machine learning algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forest have been widely used. These methods rely on manually extracted features and have shown moderate performance in identifying mental states.

In recent years, there has been a shift towards advanced approaches such as deep learning models, including Artificial Neural Networks (ANN), which can automatically learn patterns from EEG data and improve classification accuracy. These models have shown better performance compared to traditional methods, especially in handling complex and large datasets.

In the gaming domain, earlier systems relied on physical input devices, while recent research focuses on EEG-based control systems to enhance user interaction. However, many existing systems still face challenges such as noise sensitivity, reduced accuracy, and difficulty in real-time implementation.

Therefore, there is a need for improved systems that can effectively process EEG signals and provide accurate real-time predictions for practical applications.

2.1 EXISTING SYSTEM

The existing system is based on basic EEG signal acquisition and processing for simple control applications. It collects raw EEG data from a sensor through a serial interface and stores it for further analysis. The system applies standard signal

processing techniques such as notch filtering and bandpass filtering to remove noise and unwanted frequency components.

Feature extraction in the existing system mainly focuses on frequency-based characteristics such as power spectral density (PSD) across different brainwave bands (alpha, beta, theta, delta). Additional features like spectral centroid and peak frequency are also calculated. These features are then used as input to a machine learning model, which is trained to classify different brain signal patterns.

The system performs predictions in real time using the trained model. Based on the predicted output, simple control actions such as key presses are triggered using automation tools. The system is capable of detecting certain actions like jaw clenching and basic cognitive responses.

However, the overall implementation is limited in terms of dataset structure, feature diversity, and robustness in real-time performance.

2.2 DISADVANTAGES OF EXISTING SYSTEM

- a) Limited dataset collection without proper labeling for different mental states.
- b) Heavy dependence on frequency-domain features, missing time-domain analysis.
- c) Lower accuracy due to limited feature representation.
- d) Lack of structured data collection across multiple users.
- e) No proper handling of data imbalance.
- f) Less robust real-time performance due to signal variability.
- g) Limited preprocessing techniques for noise reduction.
- h) Not specifically designed for cognitive state classification (focus vs relaxation).
- i) Less optimized control mechanism, leading to unstable outputs.

2.3 PROPOSED SYSTEM

The proposed system improves upon the existing approach by developing a structured and efficient cognitive gaming control system using EEG signals. It introduces a systematic method for data collection, where EEG signals are recorded for specific mental states such as attentive and relaxed under controlled conditions.

The system incorporates improved preprocessing techniques, including smoothing, normalization, and filtering, to enhance signal quality. It focuses on extracting both time-domain and statistical features such as mean, variance, range, energy, and peak values, which better represent the characteristics of EEG signals.

A deep learning model based on Artificial Neural Networks (ANN) is trained using the extracted features to achieve higher classification accuracy. The dataset is balanced and properly prepared to improve model performance and generalization.

In real-time implementation, the system processes incoming EEG data in segments, applies preprocessing, extracts features, and predicts the user's mental state continuously. A smoothing mechanism is used to stabilize predictions. Based on the predicted state, control actions are triggered to operate a gaming application, enabling hands-free interaction.

2.4 ADVANTAGES OF PROPOSED SYSTEM

- a) Structured and labeled dataset collection for better learning
- b) Improved feature extraction using both statistical and signal-based features
- c) Higher accuracy using deep learning (ANN) model
- d) Better noise handling through advanced preprocessing techniques
- e) Balanced dataset for improved model performance
- f) Stable and reliable real-time prediction using smoothing techniques
- g) Efficient control mechanism for gaming applications
- h) Enhanced user interaction through cognitive state detection
- i) More adaptable and scalable for future improvements

3. ANALYSIS

The analysis of this project focuses on understanding how EEG signals can be effectively used to detect a user's cognitive state and control a gaming application in real time. Since EEG signals are highly sensitive and vary between individuals, it is important to analyze the challenges involved in signal acquisition, processing, and classification.

In this project, analysis is carried out on how to collect reliable EEG data for different mental states such as attentive and relaxed. It also involves studying how noise and unwanted disturbances affect the signal and identifying suitable preprocessing techniques to improve signal quality.

Another important aspect of analysis is determining the type of features that can best represent EEG signals. Various statistical and signal-based features are considered to ensure that the system can accurately distinguish between different cognitive states.

The project also analyzes the performance of different approaches for classification and selects a suitable model that can provide better accuracy and real-time prediction capability. Special attention is given to handling continuous data streams and ensuring stable output during real-time operation.

Overall, this phase focuses on identifying the key requirements and challenges involved in building a reliable EEG-based cognitive control system and defining the approach to achieve accurate and efficient performance.

3.1 SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirement Specification for this project focuses on the functionalities needed to process EEG signals and control a gaming application in real time.

The system should be able to read EEG data continuously from the device through serial communication. It must support preprocessing techniques such as filtering and normalization to improve signal quality.

The software should extract meaningful features from the EEG data and use a trained model to classify the user's mental state as attentive or relaxed. It should also support real-time prediction and provide stable output.

Finally, the system must be able to trigger game control actions based on the predicted results, ensuring smooth and responsive interaction.

3.1.1 USER REQUIREMENTS

The user of the system should be able to easily interact with the EEG-based cognitive gaming system without requiring advanced technical knowledge. The system should provide a simple and user-friendly setup process, allowing the user to wear the EEG device correctly and start the system with minimal effort.

The user should be able to perform different mental states, such as attentive (focused) and relaxed, which are required for both training and real-time prediction. The system should clearly guide the user during data collection and ensure that the process is smooth and understandable.

During operation, the system should work in real time and respond quickly to the user's mental state. The user should experience smooth game control without noticeable delay or interruptions.

The system should also ensure basic comfort and safety while using the EEG device, including proper placement of electrodes and stable signal acquisition. Additionally, the system should provide reliable and consistent performance so that the user can effectively control the game.

3.1.2 SOFTWARE REQUIREMENTS

Software Requirements:

The system is developed using Python due to its strong support for data processing and machine learning. Libraries such as NumPy and Pandas are used for handling and analyzing data, while SciPy is used for signal processing operations like filtering. Scikit-learn is used for preprocessing and utility functions, and TensorFlow/Keras is used to build and train the deep learning model.

PySerial enables communication between the EEG hardware and the system, allowing real-time data acquisition. PyAutoGUI is used to automate game controls based on predictions. Joblib is used for saving and loading the trained model and scaler.

The development and execution of the system are carried out using tools like VS Code or Jupyter Notebook.

S.No	Software / Tool	Description / Purpose
1	Operating System (Windows)	Platform for running the system
2	Python	Core programming language used for development
3	NumPy	Used for numerical computations and array operations
4	Pandas	Used for data handling and dataset processing
5	SciPy	Used for signal processing operations
6	Scikit-learn	Used for preprocessing and data handling utilities
7	TensorFlow / Keras	Used for building and training the model
8	PySerial	Used for communication with Arduino
9	PyAutoGUI	Used for controlling game actions
10	Joblib	Used for saving and loading model components
11	VS Code / Jupyter Notebook	Development and execution environment

Table. 3.1.2 software requirements

3.1.3 HARDWARE REQUIREMENTS

Hardware Requirements:

The EEG sensor system consists of electrodes placed on the scalp to capture brain signals. These electrodes detect very small electrical signals generated by brain activity. The signals are then passed to a BIO AMP (EXG pill), which amplifies the weak EEG signals to make them suitable for further processing.

The amplified signals are transmitted to an Arduino or similar microcontroller, which converts the signals into digital data and sends them to the computer through serial communication.

A computer or laptop is used to run the software, process the EEG data, and perform classification using the trained model. A USB cable is used to connect the hardware components and ensure proper data transfer between the device and the system.

S.No	Hardware Component	Description / Purpose
1	EEG Sensor Device	Captures brain signals from the user
2	Electrodes	Used for acquiring signals from the scalp
3	BIO AMP / EXG Pill	Amplifies the weak EEG signals
4	Arduino (Microcontroller)	Converts analog signals to digital data
5	Computer / Laptop	Used for processing and running the system
6	USB Cable	Connects Arduino to the computer

Table. 3.1.3 hardware requirements

3.2 CONTENT DIAGRAM OF THE PROJECT

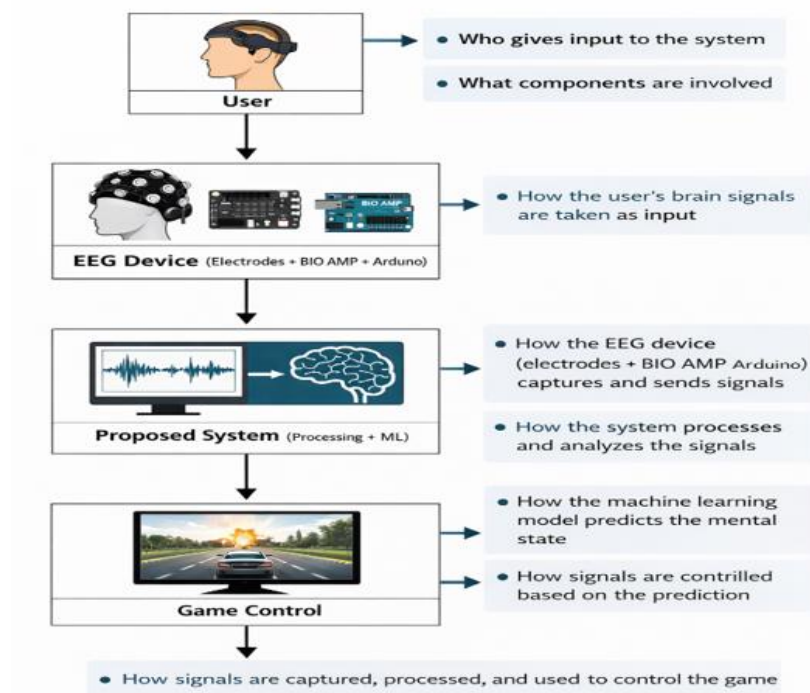


Fig.3.2 content diagram

The content diagram illustrates the overall flow of the EEG-based cognitive gaming control system. It begins with the user, who provides input in the form of brain signals. These signals are captured using the EEG device, which includes electrodes, a BIO AMP, and an Arduino for signal acquisition. The captured signals are then sent to the proposed system, where they are processed and analyzed using signal processing and machine learning techniques. The system extracts meaningful information and predicts the user's mental state. Based on this prediction, appropriate control actions are generated in the game control module. Thus, the diagram clearly shows how brain signals are captured, processed, and converted into game control actions, representing the complete working flow of the system.

4. DESIGN

4.1 INTRODUCTION

The design phase of this project focuses on structuring and organizing the components required to develop the EEG-based cognitive gaming control system. It defines how the system is built to handle real-time brain signal processing and convert it into meaningful control actions for a gaming application.

In this project, the design is centered around a pipeline approach, where EEG signals flow through multiple stages starting from data acquisition to final game control. The system is designed to efficiently handle continuous data input from the EEG device, which includes electrodes, BIO AMP (EXG pill), and a microcontroller for signal transmission.

A key aspect of the design is ensuring proper preprocessing of EEG signals, as they are highly sensitive to noise and external disturbances. Therefore, filtering and normalization techniques are incorporated to improve signal quality. The design also includes feature extraction methods that convert raw signals into meaningful numerical representations, making them suitable for classification.

Another important part of the design is the integration of machine learning and deep learning models. The system is designed to use extracted features as input to the trained model, which predicts the user's mental state in real time. To improve reliability, the design includes mechanisms for handling continuous data streams and stabilizing predictions.

Finally, the system is designed to generate control signals based on the predicted output and interact with a gaming application. The overall design ensures smooth data flow, efficient processing, and real-time response, making the system effective for cognitive gaming control.

4.2 USECASE DIAGRAM / DFD / CLASS DIAGRAM

4.2.1 USE CASE DIAGRAM

The use case diagram shows how the user interacts with the EEG-based cognitive gaming system. The user first wears the EEG device and starts data collection, where brain signals are recorded and stored. These signals are used to train the model.

For real-time operation, the user starts the prediction system. The system processes EEG signals, extracts features, and predicts the mental state as attentive or relaxed. Based on the prediction, the system controls the game accordingly.

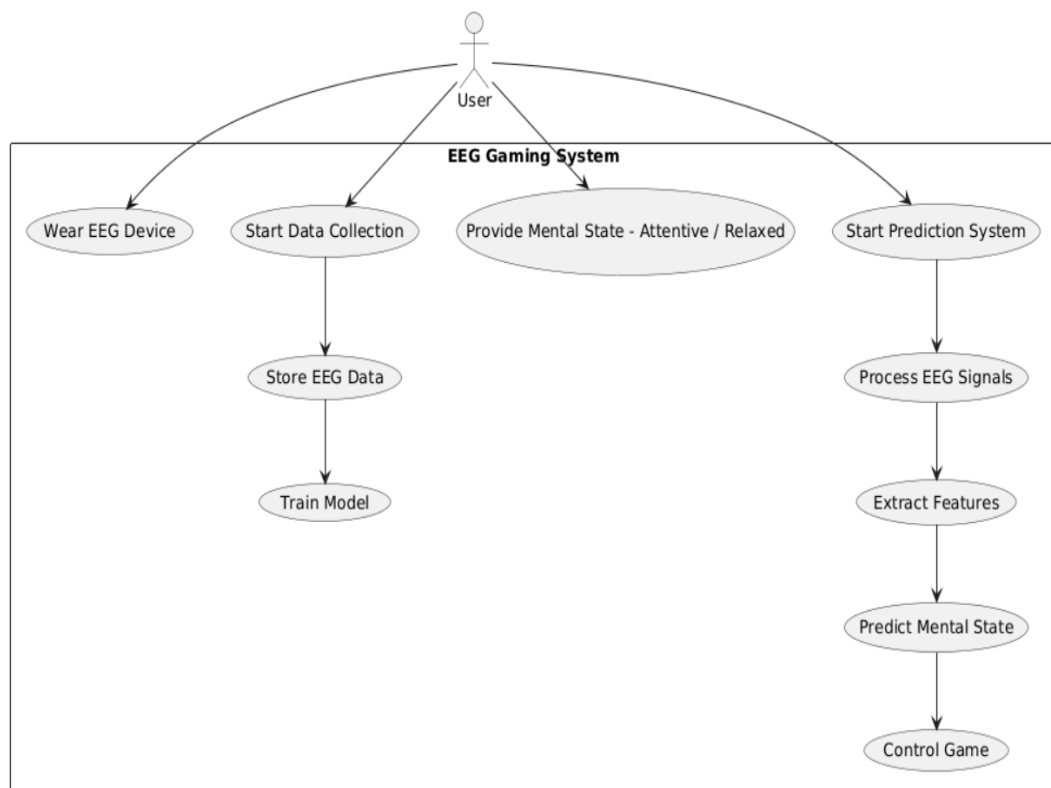


Fig-4.2.1 use case diagram

4.2.2 DATA FLOW DIAGRAM (DFD)

The DFD shows how EEG data flows through the system. The user generates brain signals, which are captured by the EEG device and sent as raw data. The signals are then preprocessed to remove noise and converted into clean signals.

Next, features are extracted and given to the ML/DL model for classification. The model predicts the mental state as attentive or relaxed. Based on this prediction, the game control module performs actions such as moving or stopping the game.

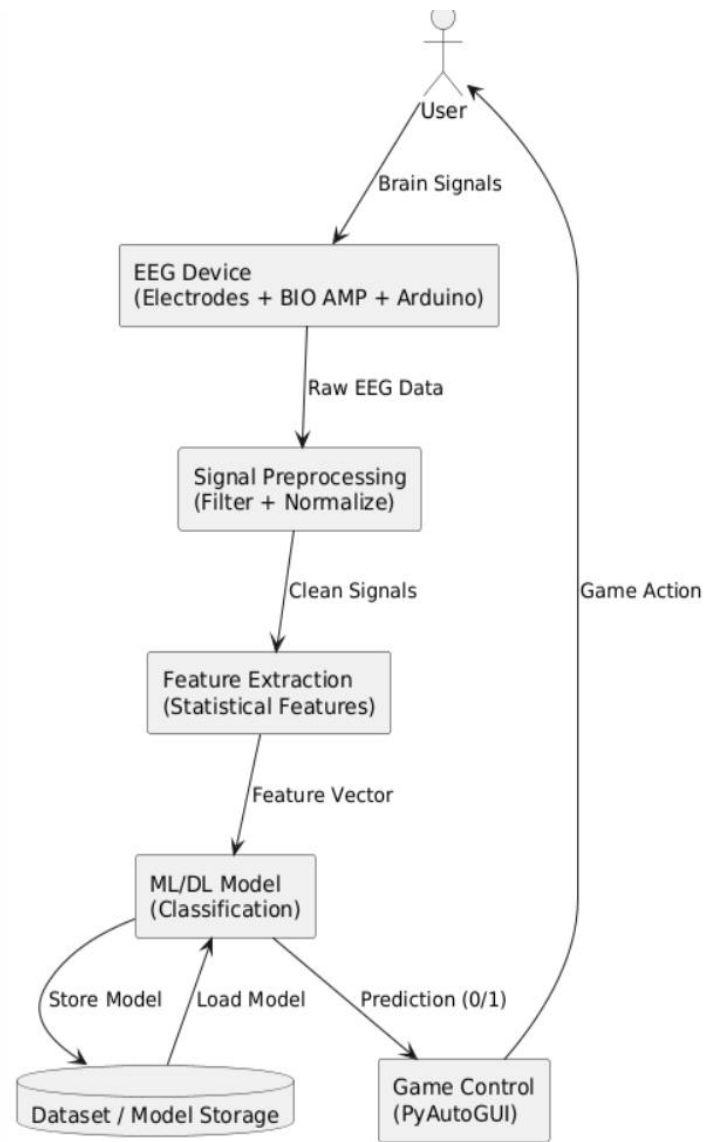


Fig. 4.2.2 data flow diagram

4.2.3 CLASS DIAGRAM

The class diagram represents the structure of the EEG-based cognitive gaming control system and the interaction between its main components.

The User class generates brain signals, which are captured by the EEG_Device class. The EEG device is responsible for acquiring the signals and sending raw data to the system. The Preprocessing class processes the raw signals by applying filtering and normalization techniques to remove noise and improve signal quality.

The processed data is then passed to the FeatureExtraction class, where important features are extracted from the signals. These features are used by the Model class, which performs training and prediction to identify the user's mental state.

Finally, the GameController class uses the prediction results to control the game. The system also provides feedback to the user based on the game actions. This diagram clearly shows the flow of data and the responsibilities of each component in the system.

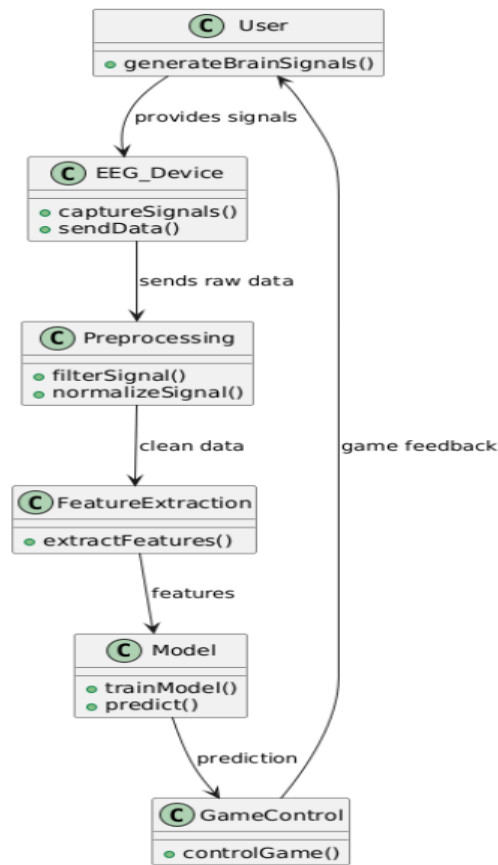


Fig. 4.2.3 class diagram

4.3 DATA DICTIONARY

The data dictionary describes the important data elements used in the EEG-based cognitive gaming control system. It helps in understanding the type, structure, and purpose of data used at different stages of the system.

EEG Signal Data

The EEG signal data represents the raw brain signals captured from the EEG device. These values are stored as floating-point numbers and are directly received from the electrodes through the BIO AMP and Arduino. This data serves as the primary input for further processing in the system.

Label Data

The label data indicates the mental state of the user and is represented as integer values. A value of 1 denotes an attentive (focused) state, while 0 represents a relaxed

state. These labels are assigned during data collection and are mainly used for training and evaluating the model.

Preprocessed Signal

The preprocessed signal refers to the cleaned version of the raw EEG data after applying filtering and normalization techniques. This step removes noise and improves signal quality, making it suitable for feature extraction.

Feature Vector

The feature vector consists of important statistical features extracted from the EEG signals, such as mean, standard deviation, variance, energy, and peaks. These features summarize the signal information and are used as input for the machine learning model.

Model Input Data

The model input data represents a collection of feature vectors arranged in a structured format. This data is used for training and testing the machine learning or deep learning model.

Model Output Data

The model output data represents the predicted mental state of the user, which is either attentive or relaxed. This output is generated by the classification model based on the input features.

Scaled Features

Scaled features are the normalized version of the extracted features. Scaling ensures that all feature values are within a similar range, which helps in improving the performance and accuracy of the model.

Control Signal

The control signal is the final output generated by the system based on the predicted mental state. It is used to perform actions in the game, such as moving or stopping, using automation tools.

Dataset File

The dataset file contains the collected EEG signal values along with their corresponding labels. It is stored in CSV format and is used for training the model.

Model File

The model file stores the trained machine learning or deep learning model. It is used during the prediction phase to classify the user's mental state based on new input data.

S.No	Data Name	Data Type	Description
1	value	Float	Raw EEG signal value captured from device
2	label	Integer (0/1)	Represents mental state (0 = Relaxed, 1 = Attentive)
3	processed_signal	Array (Float)	Signal after preprocessing (noise removed)
4	features	Array	Extracted features from signal
5	X	2D Array	Input data used for model training/testing
6	y / prediction	Integer	Output representing predicted mental state
7	scaled_features	Array	Normalized feature values
8	control_signal	Action/Command	Output action (Move/Stop)
9	final_dataset.csv	File	Dataset containing values and labels
10	brain_model.h5	File	Trained model file

Table. 4.3 data dictionary

5. IMPLEMENTATION & RESULTS

5.1 METHOD OF IMPLEMENTATION

Data Collection Process

The implementation of the BCI system begins with the collection of EEG brain signals using a sensor connected through serial communication. The system reads real-time signal values from the sensor and stores them along with corresponding labels indicating the user's mental state. The data collection is performed in multiple cycles, where each cycle consists of attentive (focused) and relaxed states. This structured approach ensures that sufficient and balanced data is collected for model training.

Dataset Preparation

After collecting data from multiple users, the datasets are combined into a single dataset for further processing. The system merges all individual datasets into a unified file, ensuring consistency in format and structure. The final dataset contains two main columns: signal values and their corresponding labels. This step is crucial for creating a comprehensive dataset suitable for machine learning.

Data Preprocessing and Segmentation

The collected EEG data is preprocessed to improve its quality. The system applies normalization and segmentation techniques, where the signal is divided into fixed-size windows. Each segment represents a portion of the brain signal that can be analyzed independently. This step ensures that the data is properly structured for feature extraction and model training.

Feature Extraction

From each segmented signal, meaningful statistical features are extracted. These include mean, standard deviation, variance, maximum and minimum values, signal range, energy, and peak count. These features help in capturing the characteristics of brain signals and act as input for the deep learning model. Feature extraction simplifies raw data into a form that is easier for the model to understand.

Model Training

A deep learning model is implemented using a neural network architecture. The model consists of multiple dense layers with activation functions to learn complex patterns in the data. The extracted features are normalized using a scaler and then used

to train the model. The training process includes splitting the dataset into training and testing sets, optimizing the model using an appropriate optimizer, and evaluating performance based on accuracy.

Real-Time Prediction and Control

In the final stage, the trained model is used for real-time prediction. The system continuously reads EEG signals, processes them, extracts features, and predicts the user's mental state. Based on the prediction, control actions are executed using keyboard simulation. When the user is focused, the system triggers forward movement, and when relaxed, it stops the movement. This completes the real-time BCI control loop.

5.1.1 IMPLEMENTATION OF ALGORITHMS

Signal Preprocessing Algorithm

```
def moving_average(signal, window=5):
    return np.convolve(signal, np.ones(window)/window, mode='same')

def low_pass_filter(signal, cutoff=3, fs=50, order=3):
    nyquist = 0.5 * fs
    normal_cutoff = cutoff / nyquist
    b, a = butter(order, normal_cutoff, btype='low')
    return filtfilt(b, a, signal)
```

Moving average smoothing reduces random fluctuations in EEG signals, while the low-pass filter removes high-frequency noise. This improves signal quality before further processing.

These methods are simple, efficient, and suitable for real-time systems compared to complex filters, making them ideal for this project.

Feature Extraction Algorithm

```
mean = np.mean(segment)
std = np.std(segment)
var = np.var(segment)
energy = np.sum(segment**2)
peaks = np.sum(segment > (mean + std))
```

Statistical features are extracted from EEG signals to represent their behavior in a numerical form.

Statistical features are computationally fast and effective, making them better suited than complex transformations for real-time prediction.

Dataset Balancing Algorithm

```
X_attentive_up, y_attentive_up = resample(  
    X_attentive, y_attentive,  
    replace=True,  
    n_samples=len(X_relaxed),  
    random_state=42  
)
```

Upsampling is used to balance the dataset by increasing the number of samples in the minority class.

It prevents model bias and improves accuracy compared to using imbalanced data.

Deep Learning Classification Algorithm

```
model = Sequential()  
model.add(Dense(256, activation="relu", input_shape=(10,)))  
model.add(Dense(128, activation="relu"))  
model.add(Dense(64, activation="relu"))  
model.add(Dropout(0.3))  
model.add(Dense(32, activation="relu"))  
model.add(Dense(2, activation="softmax"))
```

A deep neural network is used to learn patterns from extracted features and classify mental states.

Deep learning provides better accuracy and pattern recognition compared to basic machine learning models

Real-Time Prediction Algorithm

```
if len(buffer) >= WINDOW:  
    segment = np.array(buffer[-WINDOW:])
```

A sliding window is used to process continuous EEG signals in fixed-size segments.

This approach enables real-time processing instead of waiting for complete data.

Control Action Algorithm

If $\text{avg_prob} > \text{THRESHOLD}$:

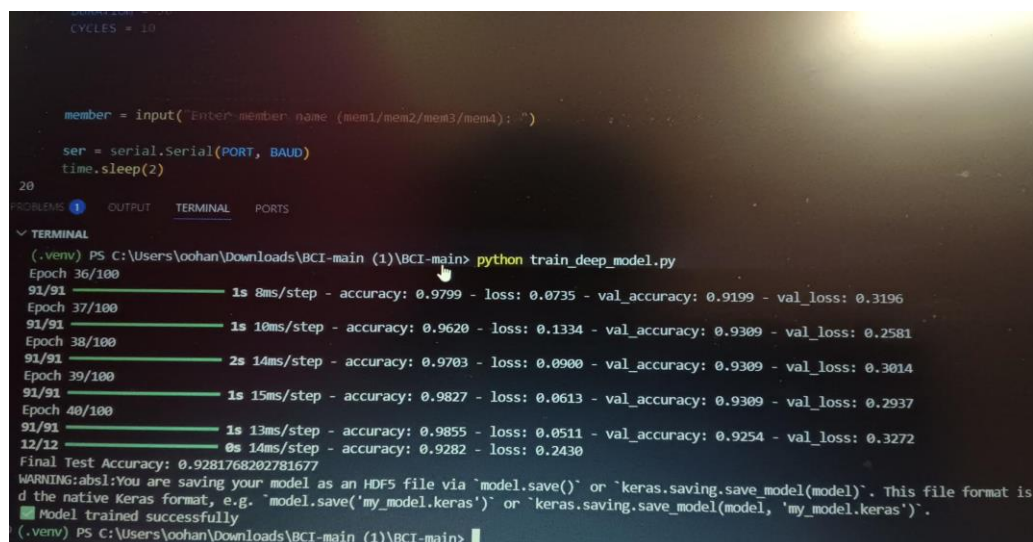
```
pyautogui.keyDown("w")
```

else:

```
pyautogui.keyUp("w")
```

Predicted output is converted into game control actions. This method ensures smooth and immediate response, making the system interactive.

5.1.2 OUTPUT SCREENS



```
CYCLES = 10

member = input("Enter member name (mem1/mem2/mem3/mem4): ")

ser = serial.Serial(PORT, BAUD)
time.sleep(2)

20

PROBLEMS 1 OUTPUT TERMINAL PORTS

▼ TERMINAL
(.venv) PS C:\Users\oohan\Downloads\BCI-main (1)\BCI-main> python train_deep_model.py
Epoch 36/100
91/91 1s 8ms/step - accuracy: 0.9799 - loss: 0.0735 - val_accuracy: 0.9199 - val_loss: 0.3196
Epoch 37/100
91/91 1s 10ms/step - accuracy: 0.9620 - loss: 0.1334 - val_accuracy: 0.9309 - val_loss: 0.2581
Epoch 38/100
91/91 2s 14ms/step - accuracy: 0.9703 - loss: 0.0900 - val_accuracy: 0.9309 - val_loss: 0.3014
Epoch 39/100
91/91 1s 15ms/step - accuracy: 0.9827 - loss: 0.0613 - val_accuracy: 0.9309 - val_loss: 0.2937
Epoch 40/100
91/91 1s 13ms/step - accuracy: 0.9855 - loss: 0.0511 - val_accuracy: 0.9254 - val_loss: 0.3272
12/12 0s 14ms/step - accuracy: 0.9282 - loss: 0.2430
Final Test Accuracy: 0.9281768202781677
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is
d the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`.
Model trained successfully
(.venv) PS C:\Users\oohan\Downloads\BCI-main (1)\BCI-main>
```

Fig. 5.1.2.1 accuracy score

```
CYCLES = 10

member = input("Enter member name (mem1/mem2/mem3/mem4): ")

ser = serial.Serial(PORT, BAUD)
time.sleep(2)

20
PROBLEMS 1 OUTPUT TERMINAL PORTS
v TERMINAL
(.venv) PS C:\Users\oohan\Downloads\BCI-main (1)\BCI-main> python prediction.py
Raw: 504.0
Prediction: [[1. 0.]]
Focus Prob: 0.000 | Avg: 0.000
Relax → Car Stopped
Raw: 504.0
Prediction: [[1. 0.]]
Focus Prob: 0.000 | Avg: 0.000
Relax → Car Stopped
Raw: 503.0
Prediction: [[1. 0.]]
Focus Prob: 0.000 | Avg: 0.000
Relax → Car Stopped
Raw: 504.0
Prediction: [[1. 0.]]
Focus Prob: 0.000 | Avg: 0.000
```

Fig. 5.1.2.2 prediction window

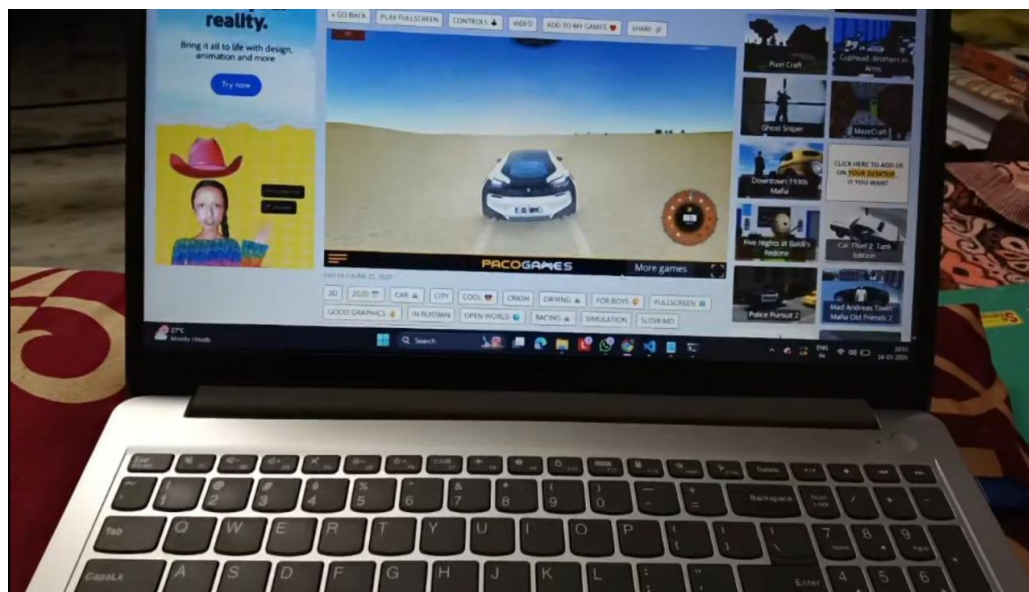


Fig. 5.1.2.3 car accelerating

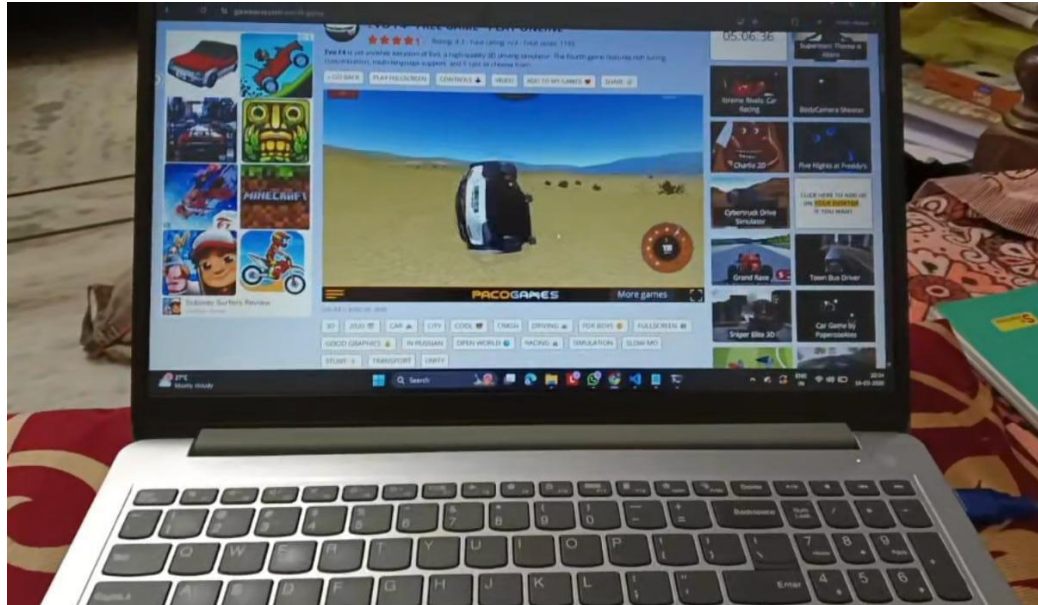


Fig. 5.1.2.4 car stops

The system displays the captured brain signal values continuously.
 It processes the signals and determines the user's current state.
 Based on this, the system generates control actions automatically.
 The game responds accordingly by moving or stopping the car.
 This shows smooth and real-time interaction between user and system.

5.1.3 RESULT ANALYSIS

The results of the proposed EEG-based cognitive gaming control system demonstrate that the system is able to effectively capture, process, and interpret brain signals for controlling game actions. The system successfully distinguishes between attentive and relaxed mental states and generates appropriate outputs based on the detected state.

During testing, the system showed stable and consistent performance in real-time conditions. The signal preprocessing stage effectively reduced noise, which helped in maintaining the quality of the input signals. The extracted features provided meaningful information, allowing the system to make reliable predictions.

The response of the system was smooth, with minimal delay between signal input and output action. The game control mechanism worked accurately, where the car

moved when the user was attentive and stopped when the user was relaxed. This indicates that the system is capable of providing real-time interaction.

The system also performed consistently across multiple trials and different users, showing its ability to handle variations in input signals. However, small fluctuations were observed due to external noise and changes in user concentration levels, which slightly affected prediction stability.

Overall, the results confirm that the system achieves its intended objective of enabling brain-controlled gaming. It provides a functional, interactive, and reliable solution, demonstrating the practical feasibility of using EEG signals for real-time applications.

6. TESTING & VALIDATION

6.1 FEATURES TO BE TESTED

EEG Signal Acquisition: The first and most important feature to be tested in the BCI system is the EEG signal acquisition process. In this stage, brainwave signals are captured using an EEG sensor connected through serial communication. The system must ensure continuous and accurate signal transmission without interruptions. Proper testing is required to verify that the incoming signal values are stable, correctly formatted, and free from communication errors.

Data Labeling Mechanism: The data labeling mechanism is another critical feature that must be validated. During data collection, the brain signals are categorized into attentive and relaxed states. This labeling forms the foundation for model training, and any incorrect labeling can lead to poor classification performance. Testing ensures that labels are correctly assigned based on the user's mental state during data recording.

Data Preprocessing Stage: The preprocessing stage is responsible for improving signal quality before analysis. Techniques such as moving average smoothing and low-pass filtering are applied to remove noise and unwanted variations. Testing verifies that these preprocessing methods are functioning correctly and producing stable signals suitable for further processing.

Feature Extraction Process: Feature extraction converts raw EEG signals into meaningful numerical representations. Statistical features such as mean, standard deviation, variance, energy, peak count, and signal range are calculated. Testing ensures that these features are accurately extracted and consistently represent the characteristics of the input signal.

Deep Learning Model Performance: The deep learning model is tested for its ability to classify brain signals into attentive and relaxed states. The model must demonstrate good accuracy and reliability on unseen data. Testing evaluates how well the model generalizes and whether it can correctly identify patterns in real-time conditions.

Real-Time Prediction Capability: The system must be capable of making predictions in real time. Testing ensures that incoming EEG signals are processed continuously and predictions are generated without delay. This is essential for maintaining smooth interaction between the user and the system.

System Control Mechanism: The final feature tested is the system control mechanism, where predicted outputs are converted into actions. Based on the mental state, the system controls the car movement—moving forward when focused and stopping when relaxed. Testing confirms that actions are executed correctly and instantly.

6.2 FEATURES NOT TO BE TESTED

Hardware-Level EEG Accuracy: The accuracy of the EEG sensor hardware is not tested as part of this system. It is assumed that the sensor provides correct and reliable signal data. Any hardware calibration or internal functioning is beyond the scope of this project.

Environmental Noise Factors: External environmental factors such as electrical interference and noise are not tested. The system assumes a controlled environment where signal disturbances are minimal. Handling such variations is not included in the current implementation.

Advanced Signal Analysis: Advanced EEG signal analysis techniques, such as frequency-domain analysis (alpha, beta waves), are not included in this project. The system focuses only on time-domain statistical features, and therefore these advanced methods are not tested.

Multi-Class Classification: The system is designed for binary classification (attentive vs relaxed). Multi-class classification involving multiple mental states is not implemented and hence not tested.

Cloud Integration and Storage: Cloud-based data storage, remote monitoring, and online processing are not part of the current system. Therefore, these features are not included in the testing process.

Graphical User Interface (GUI): The system does not include a graphical user interface for visualization or interaction. As a result, GUI-related testing is not performed.

6.3 TESTING TOOLS AND ENVIRONMENT

Programming Language and IDE: The system is developed using Python, which provides strong support for data processing and machine learning. Development and testing are carried out using tools such as Visual Studio Code, which offers an efficient environment for coding and debugging.

Libraries and Frameworks: Various libraries are used for implementation and testing. NumPy and Pandas are used for data handling, SciPy is used for signal processing, TensorFlow/Keras is used for deep learning model development, and Scikit-learn is used for preprocessing and scaling. PyAutoGUI is used for simulating keyboard actions, and Joblib is used for saving and loading models.

Hardware Setup: The system uses an EEG sensor connected via a serial port (COM9) for collecting brain signals. This hardware setup is tested to ensure smooth communication between the sensor and the system.

Operating System: The system is tested on a Windows operating system, which supports all required tools, libraries, and drivers for execution.

Types of Testing Performed: Different types of testing are performed to validate the system. Unit testing is used to test individual components such as data collection and preprocessing. Integration testing ensures that all modules work together correctly. Real-time testing is conducted to evaluate system performance under live conditions.

6.4 DESIGN OF TEST CASES AND SCENARIOS

Data Collection Test Case: This test case verifies whether the system correctly collects EEG signals and stores them in a dataset. The expected result is a properly formatted CSV file containing signal values and corresponding labels.

Dataset Combination Test Case: This test checks whether multiple datasets collected from different users are correctly merged into a single dataset. The output should be a unified dataset ready for training.

Feature Extraction Test Case: This scenario tests whether the system correctly extracts statistical features from EEG signals. The expected result is a feature vector representing each signal segment.

Model Training Test Case: This test case verifies whether the deep learning model is successfully trained using the prepared dataset. The model should achieve good accuracy and be saved for further use.

Real-Time Prediction Test Case: This scenario evaluates the system's ability to process live EEG signals and generate predictions continuously. The output should be stable and responsive.

System Control Test Case: This test case validates whether the system correctly converts predictions into actions. The expected result is correct car movement—forward when focused and stop when relaxed.

7. CONCLUSION & FUTURE ENHANCEMENT

7.1 CONCLUSION

The proposed EEG-based cognitive gaming control system presents an innovative and practical approach to controlling a game using brain signals. The system successfully captures EEG signals from the user through the hardware setup, processes the signals, and determines the user's mental state in real time. Based on this interpretation, appropriate control actions are generated, enabling the user to interact with the game without relying on traditional input devices such as keyboards or controllers. This demonstrates a shift from conventional interaction methods to a more natural and hands-free approach.

The system performs effectively by maintaining stable signal processing and providing smooth and responsive output actions. The ability to distinguish between attentive and relaxed states allows the system to control the game accurately, ensuring a consistent user experience. The real-time processing capability ensures that there is minimal delay between user input and system response, which is essential for interactive applications like gaming. The system also shows reliability across multiple trials, indicating that it can handle continuous input signals efficiently.

Furthermore, this project highlights the practical feasibility of integrating brain signal processing with interactive applications. It not only serves as a proof of concept for cognitive control but also opens opportunities for further development in this field. Overall, the project successfully achieves its objectives by developing a simple, functional, and effective brain-controlled gaming system, demonstrating its potential for future advancements in human-computer interaction.

7.2 FUTURE ENHANCEMENT

The proposed system can be further improved in several ways to enhance its performance, accuracy, and usability. One of the major areas of enhancement is improving the quality of EEG signal acquisition by using more advanced sensors and better noise reduction techniques. This will help in obtaining more accurate and stable signals, leading to better system performance.

The system can also be extended to support multiple control actions instead of only basic commands. By increasing the number of detectable mental states, more complex and interactive gaming environments can be developed. This would make the system more engaging and closer to real-world gaming experiences.

Another possible enhancement is the implementation of more advanced processing techniques to improve real-time performance and reduce delays. Optimizing the system for faster computation can make the interaction smoother and more responsive.

Additionally, the system can be expanded beyond gaming applications. It can be used in assistive technologies to help physically challenged individuals control devices without physical effort. It can also be integrated into smart environments, robotics, and healthcare systems for various practical applications.

Overall, future enhancements can make the system more robust, accurate, and versatile, increasing its scope and usability in real-world scenarios.

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